Syed Taha Ali

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Computer Science Graduate – LJMU-Oryx University
AI & Robotics | Data Science & ML | Data Security | Full Stack Developer | Researcher & Published Author

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SUMMARY

Computer Science graduate and published researcher with multidisciplinary experience spanning AI & ML, robotics, data science, full-stack development, data security and clinical research. Proven ability to design and build E2E, real-world systems. Developed AutonomousFleet AI – a novel, open-source, low-cost ROS2 Autonomous Robotics & AI platform.

SKILLS

Technical Skills: Programming (Java, Python), Robotics (ROS-2), AI & ML, Data Science & Analysis (MS BI Stack & IBM SPSS), Full-stack Development (JS, FastAPI), Database Management (SQL), Data Security (MS Purview), and Clinical Research.
Soft Skills: Problem-Solving, Teamwork, Communication, Leadership, Research Writing & Presentation, and Mentorship.

UNIVERSITY EDUCATION

Computer Science BSc (Hons), Liverpool John Moores University – Oryx University, Doha, Qatar (Sep 2024 – Jun 2026)
- **First Class Honours – GPA: 4.0**
- Transfer Credits from Swansea University (Sep 2022 – Jul 2024).

Medicine and Surgery MBBS, Medical School, Newcastle University, Newcastle, UK (Sep 2019 – Jul 2021)
- Higher Education Diploma in Medical Studies.

EXPERIENCE

Undergraduate Researcher (Part-time) **– LJMU-Oryx University,** Doha, Qatar (Sep 2024 – Present)
- Developed an open-source, low-cost, ROS2-based autonomous ground robot using RPi, with computer vision, NLP mission control, and swarm simulation, supervised by Professor Moad Idrissi – LJMU-OU Innovation Lab.

Cohort Representative (Part-time) **– LJMU-Oryx University,** Doha, Qatar (Sep 2025 – June 2026)
- Elected Cohort Rep, liaising between students and faculty to communicate feedback and represent cohort interests.

Student Mentor (Part-time) **– LJMU-Oryx University,** Doha, Qatar (Sep 2024 – June 2026)
- Supported classmates with learning coding concepts, course material, and assignment guidance.

Data Security Intern (Remote) **– Infotechtion,** London, UK (Sep 2023 – Nov 2023)
- Developed NIS-2 compliance framework and client outreach material; built NIS-2–compliant solutions in Microsoft Purview and co-presented with CTO, securing 5 new clients.

Principal Investigator (Remote) **– Weill Cornell Medicine-Qatar,** Doha, Qatar (Jan 2020 – Jul 2022)
- Led 3 studies under Dr. Mohamud Verjee; designed protocols; certified in Public Health Research Ethics (TRREE).

Co - Author & Investigator (Hybrid) **– National Institute of Cardiovascular Diseases,** Karachi, Pakistan (Jun 2018 – Sep 2022)
- Under Dr. Muhammad N. Khan, developed protocol, collected data from 200+ patients, used SPSS for analysis. Co-author of 2 published, peer-reviewed papers. Presented at GSRD ICMPH 2018.
- DOI: 10.7759/cureus.5917 and 10.47144/phj.v55i3.2207

APPLIED COMPUTING PROJECTS

AutonomousFleet AI – Senior Project, LJMU-Oryx Qatar – open-source, low-cost ROS2 Autonomous Robotics & AI Platform
- Designed and implemented a modular ROS 2 Humble autonomous robotics framework on Raspberry Pi 5, integrating sensor fusion, SLAM, localisation, autonomous navigation, and real-time robotic control.
- Developed AI-driven perception and decision-making systems using YOLOv8 computer vision and locally hosted large language models (Llama 3.1 and Qwen 2.5) for natural-language mission planning and execution.
- Built and evaluated a scalable swarm-intelligence simulation framework supporting cooperative exploration and 200-agent flocking experiments, producing a reproducible.
- Impact:
  - First open-source ROS 2 framework integrating: Sensor fusion, SLAM, Nav2, NLP control, swarm simulation
  - Lowered barrier of entry for Robotics RnD, affordable & wider-access, reproducible benchmarks.

QatarWork – Full-stack web platform
- Developed a secure, asynchronous backend using FastAPI, for multi-role marketplace with RESTful APIs.
- Implemented real-time WebSocket communication and core platform features including escrow-based transactions, user reviews, and admin moderation workflows.
- Engineered robust security mechanisms (JWT authentication, bcrypt hashing, end-to-end encryption) and validated system security using OWASP ZAP with zero high/critical vulnerabilities.

Premier League Player Analytics System – E2E Data Science Pipeline
- Designed and implemented an end-to-end data science pipeline for Fantasy Premier League analytics.
- Developed and evaluated 84 machine learning models (scikit-learn, LightGBM, XGBoost) using time-series cross-validation to predict player gameweek performance across 10 seasons (MAE ≈ 1.8–2.3 across positions).
- Built an interactive analytics dashboard, Streamlit, for real-time insights, predictions, and performance monitoring.

REFERENCES

Moad Idrissi	Hafeez Ur Rehman	Chaimae Asaad
Associate Professor, LJMU-OU	Dean of Computer Science, LJMU-OU	Assitant Professor, LJMU-OU
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